

UNDERSTANDING GRAMMAR REPRESENTATION IN THE HUMAN BRAIN VIA INFORMATION-RESTRICTED LLMS

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MOTIVATION AND BACKGROUND

How does the brain distinguish nouns and verbs?

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Fronto-Temporal Dichotomy

Damasio and Tranel (1993)

Patients with **brain lesions** doing **object** and **action** naming tasks

→ **Temporal** regions involved in the retrieval of **nouns**

→ **Frontal** regions involved in the retrieval of **verbs**

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What is driving the distinction?

How does the brain distinguish nouns and verbs? **It's complicated...**

Emergentist View

Bird et al. (2000), Vigliocco et al. (2011)

Noun / Verb distinction is not **lexical** but **semantic**

→ Nouns tend to be **more imageable** and related to **objects**

→ Verbs are more **computationally demanding**

→ Verbs are **less imageable** and related to **actions**

How does the brain distinguish nouns and verbs? **It's complicated...**

Combinatorial View

Shapiro et al. (2003)

Noun / Verb distinction driven by **computational** processes

→ Pseudowords: “*he wugged*” and “*the wugs*” represented in distinct brain areas

How does the brain distinguish nouns and verbs? **It's complicated...**

Distributed Account

Crepaldi et al. (2013)

Theory-blind meta-analysis based on 36 neuroimaging studies

→ 32 reliable **clusters**:

3 for **nouns** (R-ITG, L-AG, L-IPG)

1 for **verbs** (R-MTG)

3 both **nouns** and **verbs**; Task dependent (L-IFG, R-IFG, L-Insula)

→ No support for **Fronto-Temporal** dichotomy or **Embodied Cognition**

→ **Distributed** processing of nouns and verbs

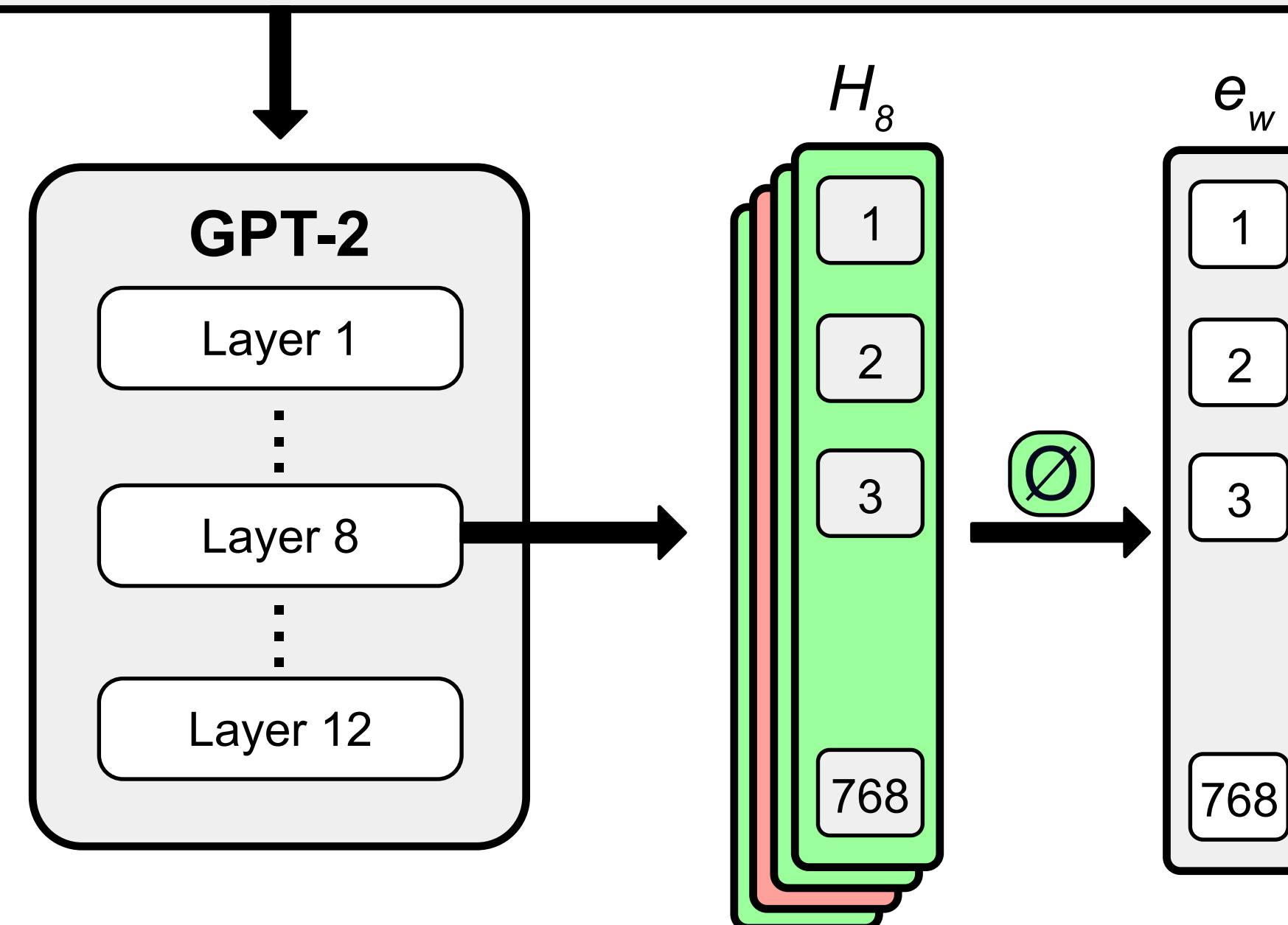
Stimulus

*There're some scientists who say that the universe
that we inhabit is not the only universe there is ...*

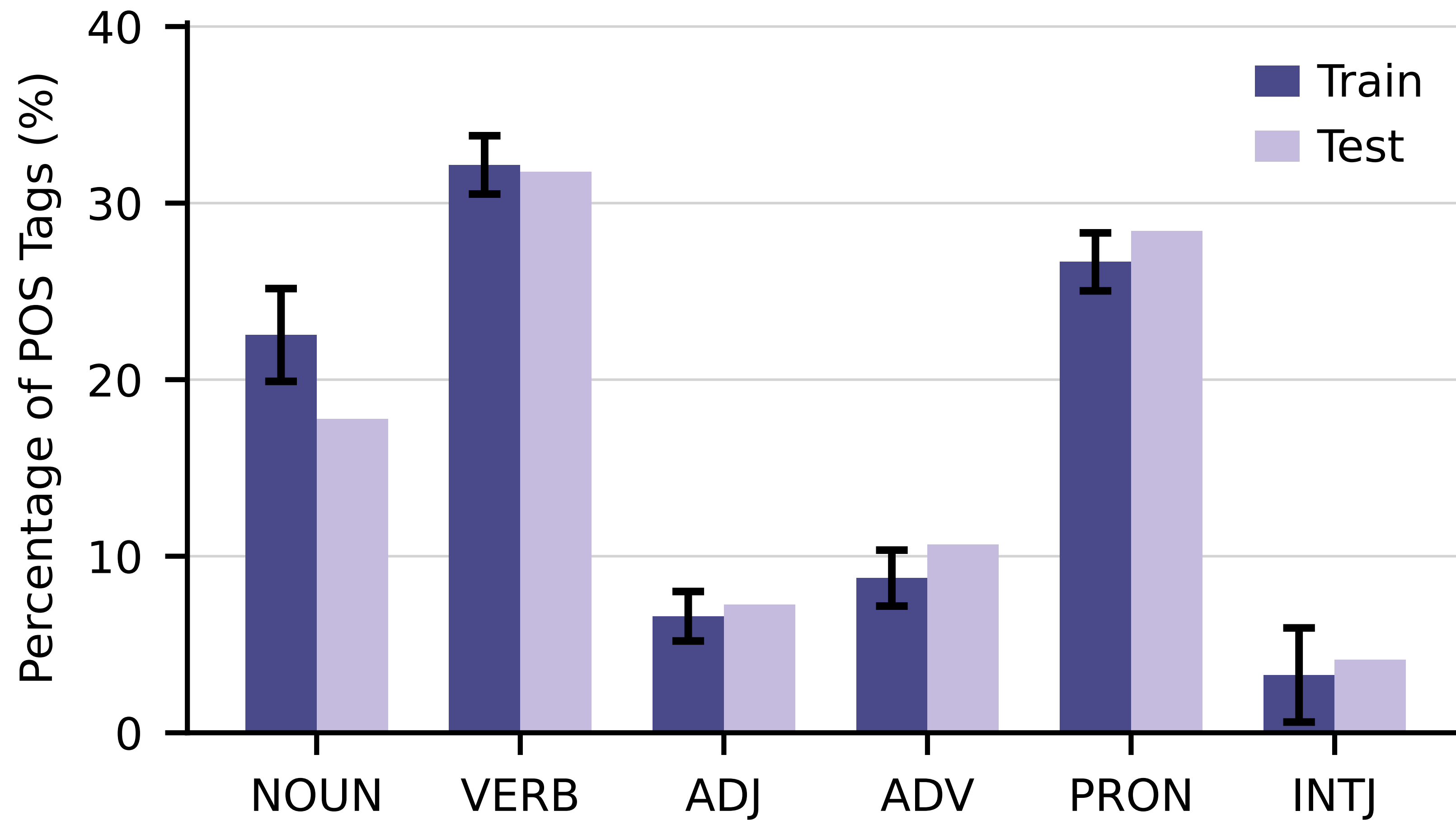
↓ POS tagging via spaCy

Attention Mask

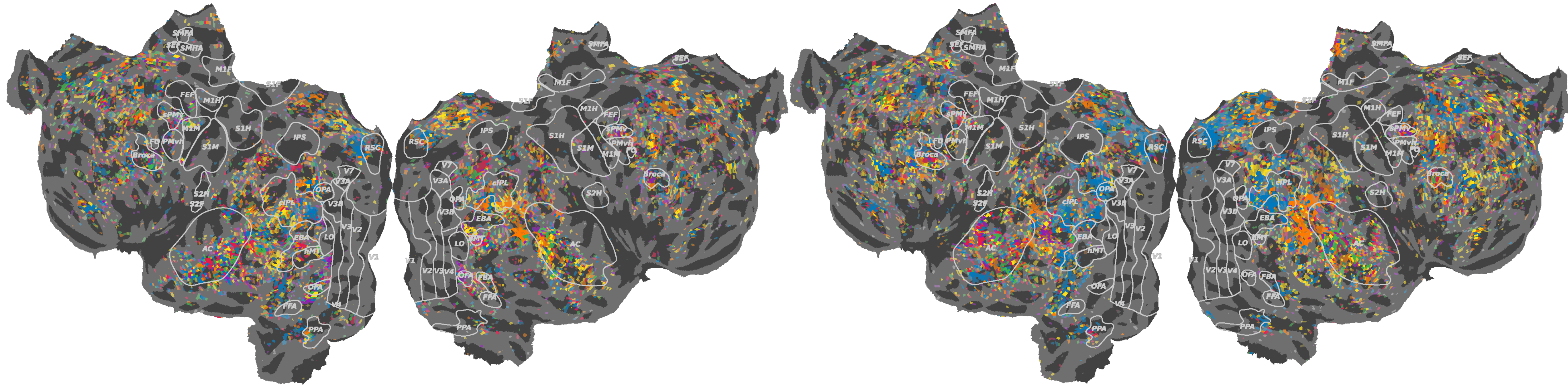
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PARTS OF SPEECH DISTRIBUTION IN DATASET



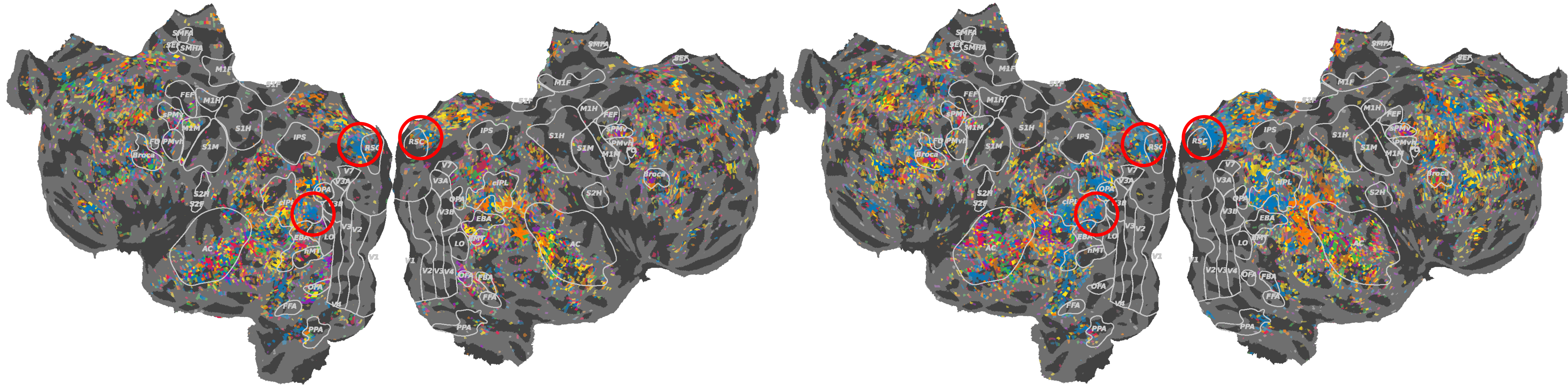
PART-OF-SPEECH SENSITIVITY



Subject 7
Reading Modality

Subject 7
Listening Modality

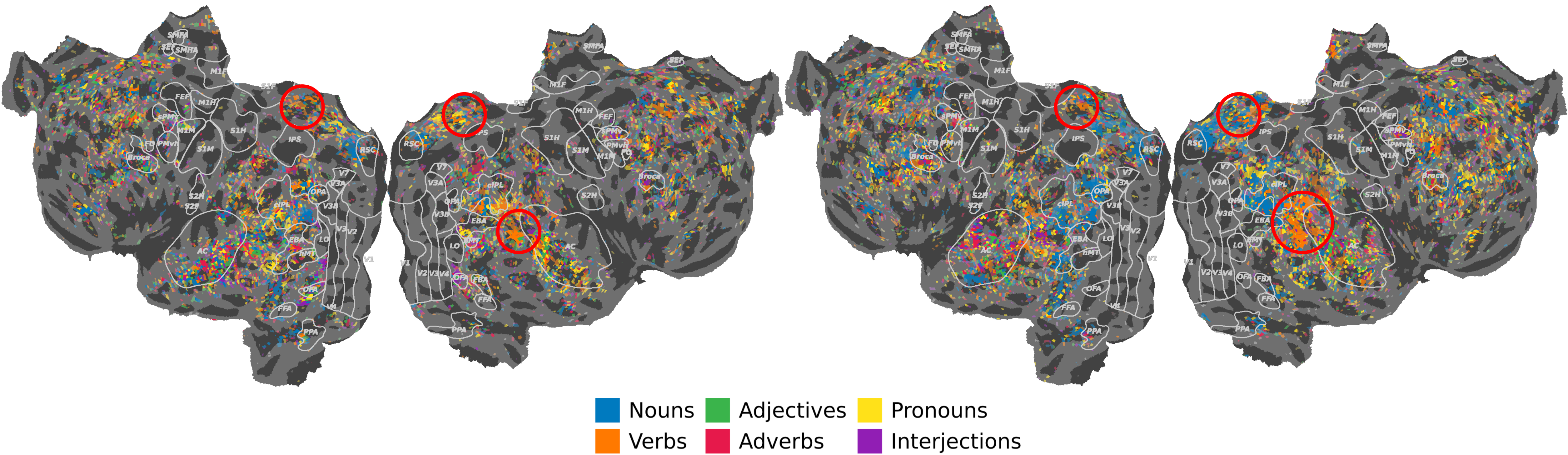
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Subject 7
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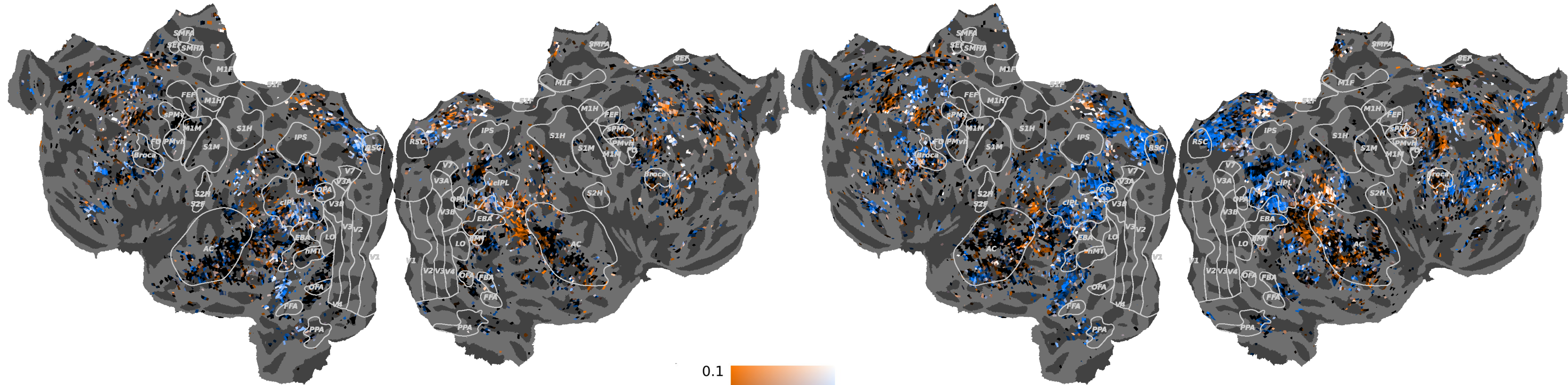
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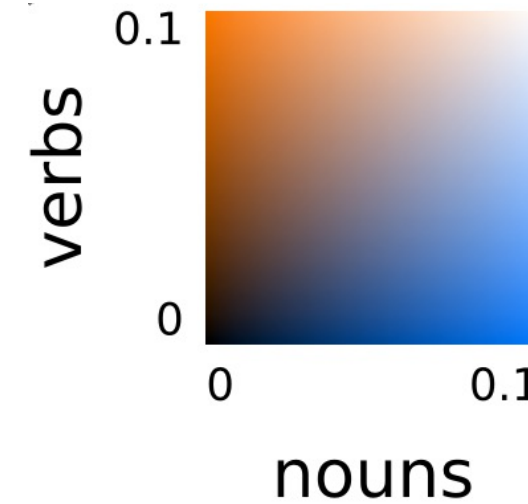
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NOUN & VERB SENSITIVITY

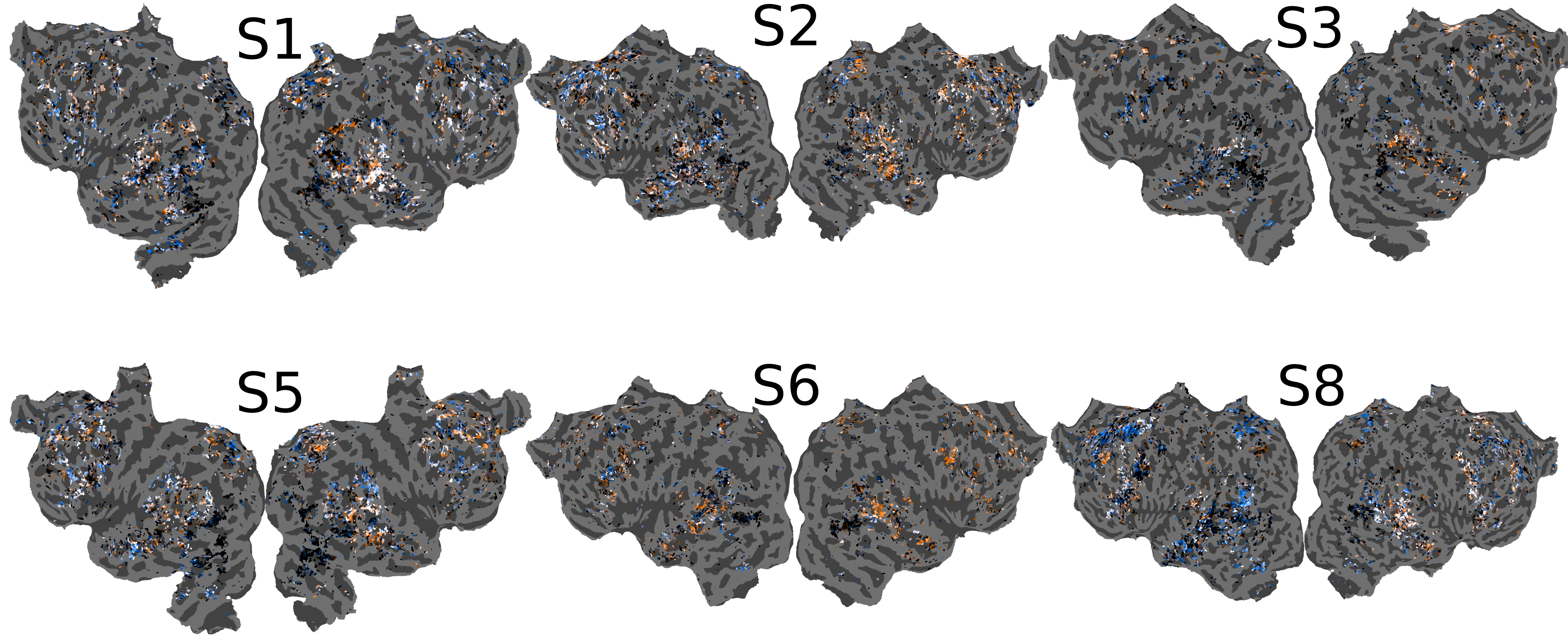


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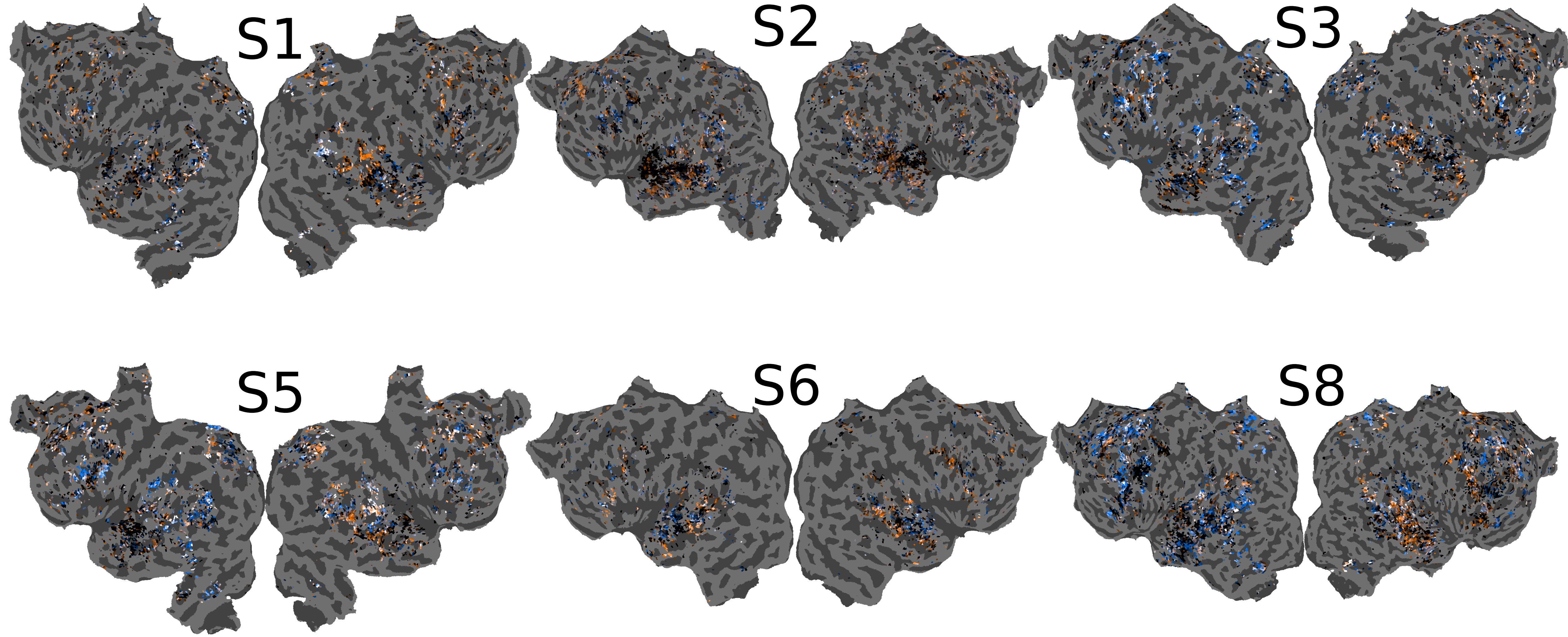
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Listening Modality



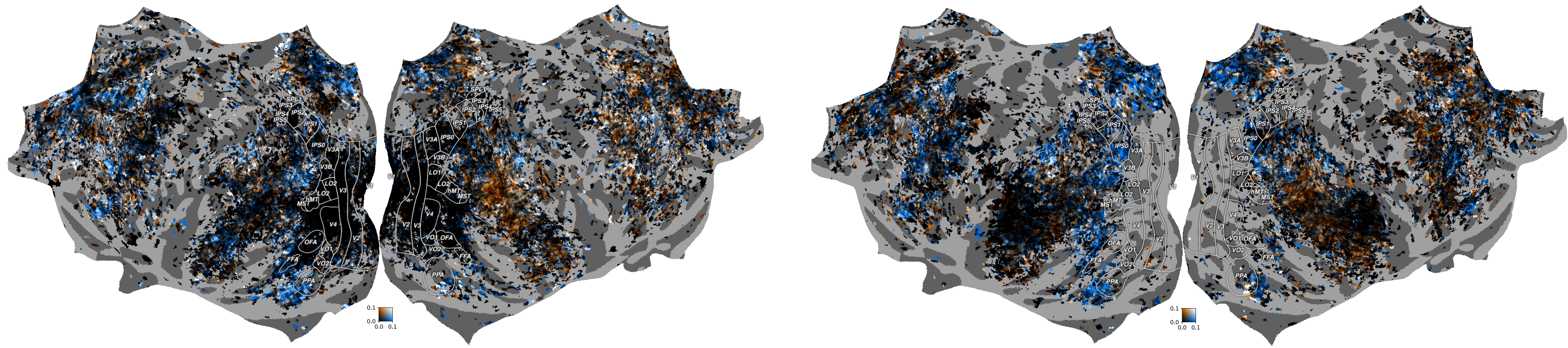
NOUN & VERB SENSITIVITY - READING



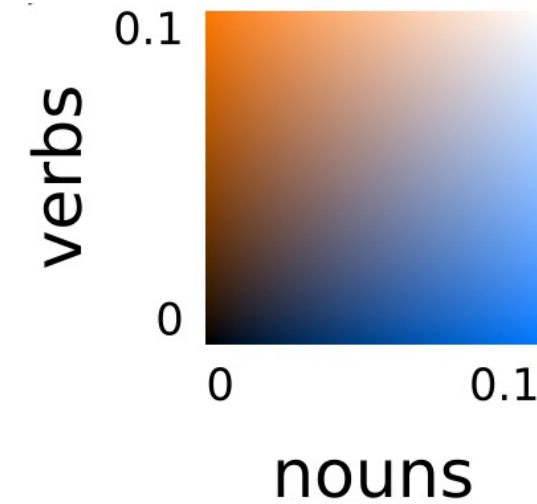
NOUN & VERB SENSITIVITY - LISTENING



NOUN & VERB SENSITIVITY ACROSS SUBJECTS

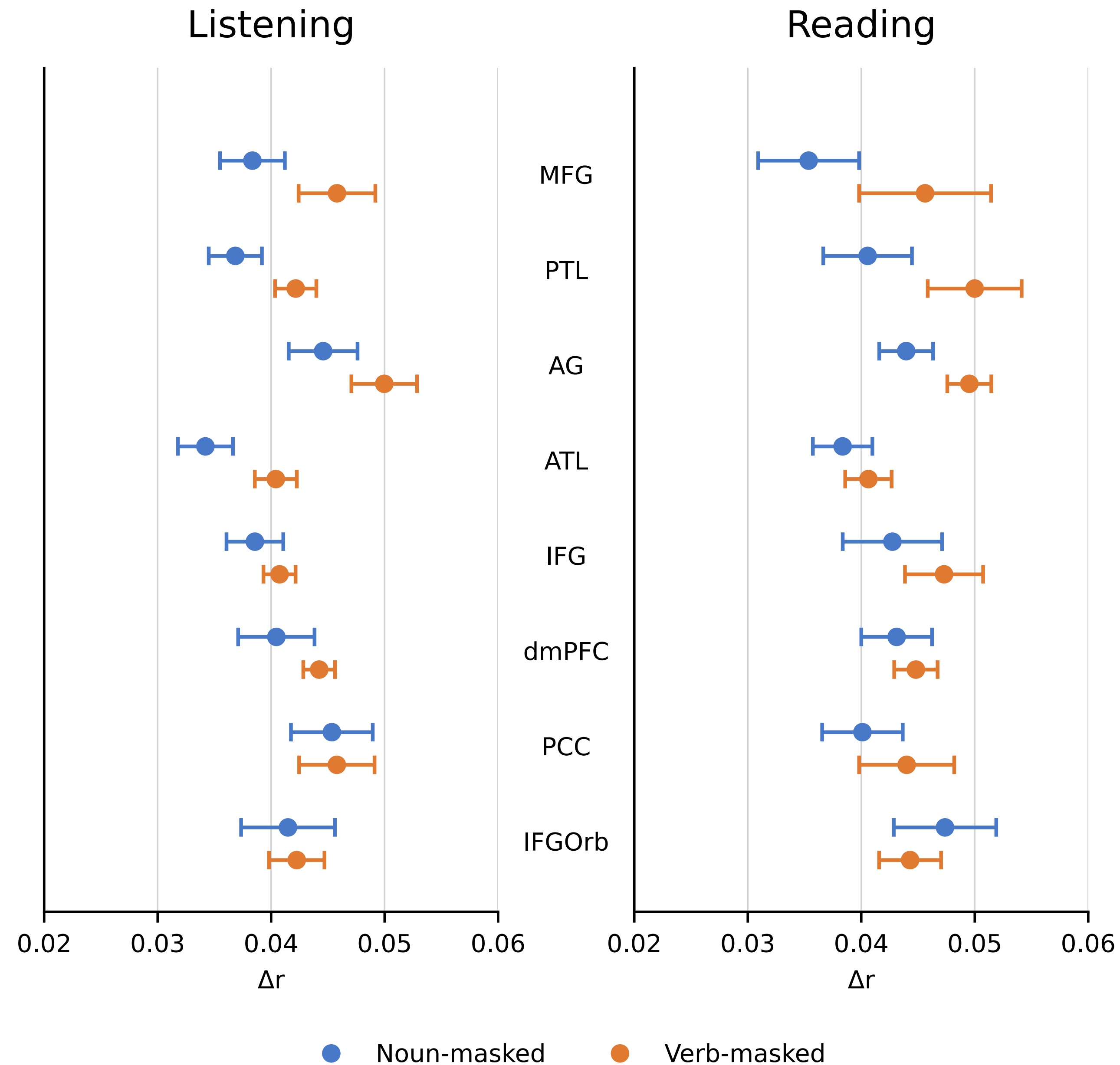


**Across Subjects
Reading Modality**



**Across Subjects
Listening Modality**

ROI ANALYSIS



Language ROIs

Fedorenko et al. (2010)

Mean Drop

0.03 – 0.06

Verb > Noun

MFG, PTL, AG, ATL

Reading vs Listening

Similar across ROIs

COMPARISON WITH LITERATURE

Theoretical account	Predicted pattern	Observed pattern
Fronto-Temporal View <i>Neuropsychological</i> (A. R. Damasio and Tranel 1993)	V > N in frontal regions; N > V in temporal regions	V > N in temporal regions (ATL, PTL); no clean frontal/temporal split.
Emergentist View <i>Object-Action, Imageability</i> (Bird et al. 2000; Vigliocco et al. 2011)	V > N in frontal and parietal regions; N > V in inferior temporal regions.	N > V near visual areas; Similar sensitivity in frontal regions.
Combinatorial <i>Morphology/inflection</i> (Tyler et al. 2004; K. A. Shapiro et al. 2005)	N = V broadly, with V > N specifically in IFG for morphological composition.	Slight V > N in IFG; N > V in IFGOrb.
Distributed <i>Shared network</i> (Crepaldi et al. 2013; Caucheteux et al. 2021)	Noun and verb circuits distributed across frontal, parietal, and temporal regions.	Smaller clusters of noun and verb sensitive voxels across the brain.

LIMITATIONS

Only the *absence* of nouns / verbs is being investigated

ROIs used in quantification don't cover all clusters /
Clusters split across multiple ROIs

Cross-Modality *not confirmed*

Investigating *presence* of nouns and verbs

Using one-hot encodings

Using a different dataset

Other than Moth

Looking at a different Language

Other than English, e.g. Chinese

Cluster Categorization

Which kind of nouns / verbs in the clusters?

Subject 7
Baseline - Verb

Subject 7
Baseline - Noun

